Gautam Manerikar

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Education

University of Toronto — Toronto, ON

Bachelor of Applied Science in Engineering Science – Robotics Engineering | *Wallberg Scholarship* — *Expected Apr 2029*

- **Relevant Coursework:** Intro to Robotics, Mathematics for Robotics, Electronics for Robotics, Dynamics, Data Structures & Analysis, Electric Circuits, Digital & Computer Systems, Differential Equations, Probability & Statistics, Thermodynamics

Experience

Autonomous Vehicle Mapping Developer — Sep 2026 – Present

aUToronto – University of Toronto — *Toronto, ON*

- Turn lidar point-cloud maps into drivable lane-level paths for the car's **20** km autonomous route across Toronto.
- Connect Lanelet2 lane segments into a routable network so the planner can find lane-level routes across the map.
- Define parent, child and sibling lanelet links so the car knows where it came from, where to go and where to merge.
- Build the HD prior map the autonomy stack relies on, offloading lane perception from live sensors onto the map.

Data Insights Analyst — May 2026 – Aug 2026

Maple Leaf Foods — *Mississauga, ON*

- 3D-scanned a worn gear with a ZEISS T-SCAN, capturing **7M+** points, and rebuilt it in CAD to identify parts to reorder.
- Owned a Power App replacing manual Excel store mapping for **18,000+** locations, iterated through nationwide user testing.
- Modelled **6M+** Power BI activity events into a star schema, delivering the first usage analytics for **1,300+** users.
- Built a PySpark pipeline in Microsoft Fabric turning review APIs into a weekly auto-updating eCommerce marketing report.

Mechanical Team Lead — Sep 2023 – Apr 2024

FIRST Robotics

- Led mechanical design and fabrication of the competition robot; placed **5th** internationally, won **2** district events.
- Developed the shooter mechanism: **2** sets of **4** flywheels linked by triangular belts, pivoting from **17°** to **55°** to set shot angle.

Projects

Gambit – Chess-Playing Robot on Bracket Bot | *Python, MuJoCo, OpenCV*

- Developed inverse kinematics and collision-checked motion planning for a **7**-DOF arm, validated in simulation to $\leq$**1 mm**.
- Implemented touch-detected placement so the arm confirms contact with the board before releasing each piece.
- Traced intermittent grasp failures to gripper pad geometry and redesigned the contact faces to eliminate them.
- Read board state via ArUco markers and a homography, inferring opponent moves by matching against all legal moves.

McMaster SumoBot Competition (**1st** Place – Division Winner) | *Arduino, Ultrasonic Sensors, IR Sensors, 3D Printing*

- Designed the full robot from chassis to control logic, integrating sensors, motors and an Arduino into one autonomous system.
- Implemented real-time opponent detection using ultrasonic and infrared sensors for distance estimation and tracking.
- Developed boundary-awareness logic to remain within the arena while maintaining aggressive engagement behavior.
- Designed and 3D printed a custom chassis, optimizing weight distribution and durability under collision forces.

PCB Christmas Tree Circuit | *LTspice, KiCad, PCB Fabrication, Soldering*

- Designed and fabricated a decorative LED Christmas tree PCB on copper-clad board, including schematic layout in KiCad.
- Simulated the circuit in LTspice to select resistor values that keep LED current safe on a **9V** battery supply.
- Etched the circuit, drilled through-holes, and soldered **8** LEDs driven by **2** transistors into a working flashing display.

Lithium-Ion Battery Charging Optimization & Degradation Simulator | *Python*

- Built a battery simulator with a team of four, tracking charge level, temperature, voltage losses and aging together.
- Compared **4** charging strategies across charging speeds and voltage limits, keeping the cell under **60°C**.
- Carried battery aging across **10** repeated charge cycles to show how each strategy wears the cell down over time.
- Modelled a **3** Ah cell as a **7**-state coupled ODE system solved with RK4, sweeping **1C–3C** and **4.2–5.0** V per strategy.

Intelligent Control for Solar Water Heating | *Fusion 360, 3D Printing, Raspberry Pi Pico*

- Designed the enclosure and custom valves in Fusion 360 for a bench-scale **3**-tank solar water heater prototype.
- Wrote the bill of materials, 3D printed all parts and wired the sensors, pump and refill valve to a Raspberry Pi Pico.
- Programmed and calibrated a weight sensor to track tank water level, opening the refill valve below a set minimum.
- Paired level control with a **40°C** temperature setpoint, heating with solar first and grid power only as backup.

Technical Skills

Languages: Python, C/C++ (Arduino), SQL, MATLAB, Swift, Verilog, RISC-V Assembly
CAD & Design: SolidWorks, AutoCAD, Fusion 360, Inventor, Onshape, SketchUp, KiCad, Altium, LTspice
Hardware: PCB Design & Bring-up, Wiring Harnesses, Sensor Integration, Soldering, 3D Printing, Oscilloscopes, Power Tools
Frameworks & Tools: ROS 2, OpenCV, MuJoCo, NumPy, URDF, Git, ZEISS T-SCAN 3D Scanning, Microsoft Excel, Microsoft Powerpoint, Microsoft Word, Microsoft Office, Power BI, Microsoft Fabric